

LABORATORY RECORD

Course: Full Stack Web Development	Course Code: 23CM4121
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1. TITLE

Design Webpage using HTML5 & CSS3

2. AIM

To design and develop a responsive static webpage utilizing HTML5 semantic layout components and CSS3 styling techniques including Flexbox, hover effects, and custom UI components.

3. OBJECTIVE

- To structure web pages using modern HTML5 semantic elements (<header>, <nav>, <section>, <article>, <aside>, <footer>).
- To apply CSS3 box-sizing, custom layouts, and flexbox models.
- To create custom styled UI components like hero sections, service cards, and sidebar panels.
- To implement interactive CSS pseudo-classes such as :hover.
- To design user-friendly form controls using stylized CSS properties.

4. THEORY

HTML5 introduced semantic tags that describe their meaning to both the browser and the developer, replacing generic div layout structures with clear landmarks. CSS3 complements HTML5 by providing advanced layout modules such as Flexbox.

- **Semantic HTML5 Layout:** Tags like <header>, <nav>, <main>, <article>, <aside>, and <footer> improve SEO, web accessibility, and structural clarity.
- **CSS Flexbox Layout:** display: flex allows elements to automatically scale, align, and distribute space within a container. Properties like justify-content, align-items, and gap control structural distribution.
- **Box Model & Universal Reset:** Using * { box-sizing: border-box; margin: 0; padding: 0; } eliminates default cross-browser spacing inconsistencies.
- **Interactive UI States:** The :hover pseudo-class adds feedback states for interactive elements such as buttons and navigation hyperlinks.

5. ALGORITHM / PROCEDURE

1. Create an HTML file named index.html.
2. Set up the document skeleton with standard <meta> viewport and title tags.
3. Add a <style> block in the head tag to encapsulate CSS3 styling rules.
4. Apply a universal CSS reset setting margin: 0, padding: 0, and box-sizing: border-box.
5. Construct the top <header> containing a website logo title and navigation links (<nav>) styled with Flexbox.
6. Build a call-to-action full-width <section class="hero"> with text headings and interactive buttons.
7. Implement a main layout container using display: flex containing a <main> flex column and an <aside> panel.
8. Inside <main>, create card components (<article class="card">) styled with CSS shadows (box-shadow), flex wrapping, and images.
9. Construct a clean contact form section (<section class="contact">) with styled text inputs, textareas, and submit buttons.
10. Add a centered structural <footer> section at the bottom.
11. Save the file and open index.html in a browser.
12. Verify responsiveness, Flexbox alignment, and interactive hover effects.

6. PROGRAM

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>HTML5 & CSS Webpage</title>

<style>
*{
    margin:0;
    padding:0;
    box-sizing:border-box;
}

body{
    font-family:Arial, sans-serif;
    background:#f4f4f4;
    line-height:1.6;
}

header{
    display: flex;
    justify-content: space-between;
    align-items: center;
```

```
    background:#677ccf;
    color: white;
    padding: 20px 40px;
}

nav a{
    color: white;
    text-decoration: none;
    margin-left: 20px;
    font-weight: bold;
}

nav a:hover{
    color: yellow;
}

.hero{
    background:#b1b389;
    color:white;
    text-align:center;
    padding:50px 20px;
}

.hero button{
    margin-top:20px;
    padding:10px 20px;
    background:white;
    color:#0d6efd;
    border:none;
    border-radius:5px;
    cursor:pointer;
}

.hero button:hover{
    background:#0d6efd;
    color:white;
}

.container{
    display:flex;
    gap:20px;
    padding:20px;
}

main{
    flex:3;
}

aside{
    flex:1;
    background:white;
    padding:20px;
    border-radius:10px;
```

```
}

.cards{
  display:flex;
  gap:20px;
  flex-wrap:wrap;
}

.card{
  flex:1;
  min-width:250px;
  background:white;
  border-radius:10px;
  padding:20px;
  text-align:center;
  box-shadow:0 0 10px rgba(0,0,0,0.1);
}

.card img{
  width:100%;
  border-radius:10px;
}

.card button{
  margin-top:10px;
  padding:10px 20px;
  background:#0d6efd;
  color:white;
  border:none;
  cursor:pointer;
}

.contact{
  background:white;
  margin:20px;
  padding:20px;
  border-radius:10px;
}

.contact input,
.contact textarea{
  width:100%;
  padding:10px;
  margin:10px 0;
}

.contact button{
  padding:10px 20px;
  background:#198754;
  color:white;
  border:none;
  cursor:pointer;
}
```

```
footer{
  background:#222;
  color:white;
  text-align:center;
  padding:15px;
  margin-top:20px;
}
</style>

</head>

<body>

<header>
  <h1>Welcome</h1>
  <nav>
    <a href="#hero">Home</a> |
    <a href="#hero">About</a> |
    <a href="#cards">Services</a> |
    <a href="#contact">Contact</a>
  </nav>
</header>

<section id="hero" class="hero">
  <h2>Welcome to Our Website</h2>
  <p>Learn HTML5 and CSS with beautiful webpage designs.</p>
  <button>Get Started</button>
</section>

<div class="container">

<main id="cards">
<section>
<h2 style="margin-bottom:20px;">Our Services</h2>

<div class="cards">

<article class="card">

<h3>Web Design</h3>
<p>Create attractive and responsive websites.</p>
<button>Read More</button>
</article>

<article class="card">

<h3>HTML5</h3>
<p>Build modern webpages using HTML5 tags.</p>
<button>Read More</button>
</article>

<article class="card">
```

```


<h3>CSS Styling</h3>
<p>Design webpages with colors, layouts, and animations.</p>
<button>Read More</button>
</article>

</div>
</section>
</main>

<aside>
<h2>Latest News</h2>
<p> ->HTML5 introduces semantic elements.</p>
<p> ->CSS makes webpages beautiful.</p>
<p> ->Responsive design improves user experience.</p>
</aside>

</div>

<section id="contact" class="contact">
<h2>Contact Us</h2>
<form>
<label>Name</label>
<input type="text" placeholder="Enter your name">

<label>Email</label>
<input type="email" placeholder="Enter your email">

<label>Message</label>
<textarea rows="5" placeholder="Enter your message"></textarea>

<button type="submit">Send Message</button>
</form>
</section>

<footer>
<p>&copy; 2026 My HTML5 Website | All Rights Reserved</p>
</footer>

</body>
</html>

```

7. CODE EXPLANATION

Code / CSS Selector	Explanation
box-sizing: border-box	Includes padding and border inside the total width/height calculation of elements.
display: flex	Turns elements into flex containers to enable dynamic layouts and spatial distribution.
justify-content: space-	Pushes the logo title to the left and navigation links to the far

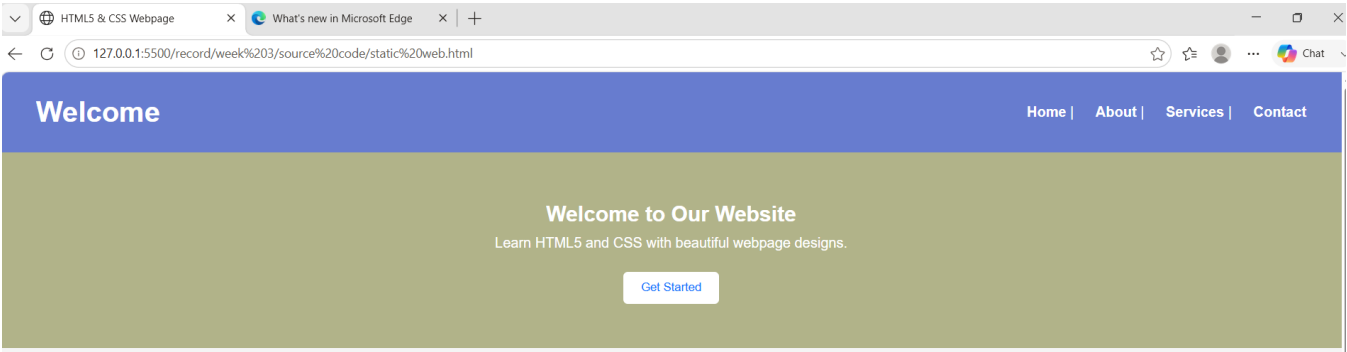
between	right.
nav a:hover	Changes anchor text color to yellow dynamically when the mouse hovers over it.
.container { display: flex }	Places <main> section (75% width via flex:3) and <aside> sidebar (25% width via flex:1) side-by-side.
flex-wrap: wrap	Allows service cards to wrap onto the next line smoothly on smaller viewport screens.
box-shadow	Applies a subtle drop shadow blur effect around card components.
<article>	HTML5 semantic element used to enclose independent card components.
<aside>	HTML5 semantic block used for secondary, related content like news widgets.

8. TEST CASES AND EXECUTION DATA

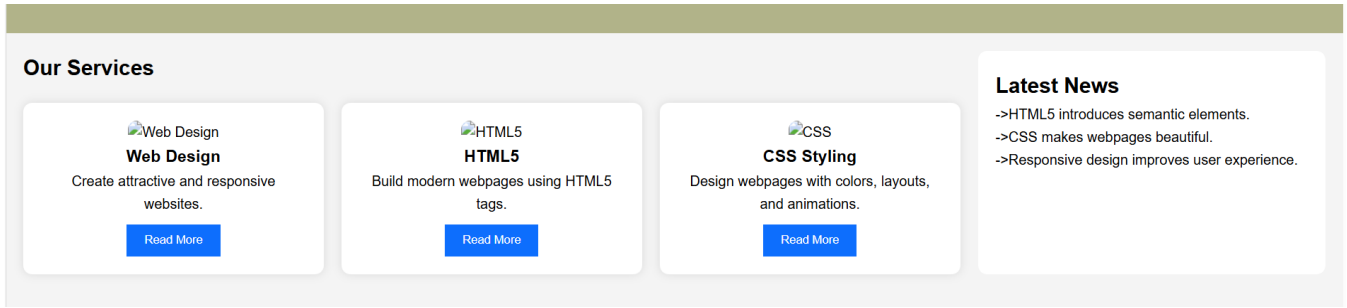
Test Case	Feature Tested	Input / Action	Expected Result	Actual Result	Status
TC-01	Navigation Anchors	Click on 'Contact' link in header	Screen scrolls down smoothly to #contact section	Scrolled down to #contact	Pass
TC-02	Hover State	Hover cursor over navigation links	Text color changes from white to yellow	Link color changes to yellow	Pass
TC-03	Flex Layout	Load webpage on full window	Main content & Sidebar sit side-by-side	Layout aligns horizontally	Pass
TC-04	Responsive Wrapping	Shrink browser window width	Cards drop down onto new lines cleanly	Cards wrap cleanly	Pass
TC-05	Form Elements	Click and type inside Name & Email fields	Keyboard input accepts text cleanly	Text input rendered	Pass

9. OUTPUT

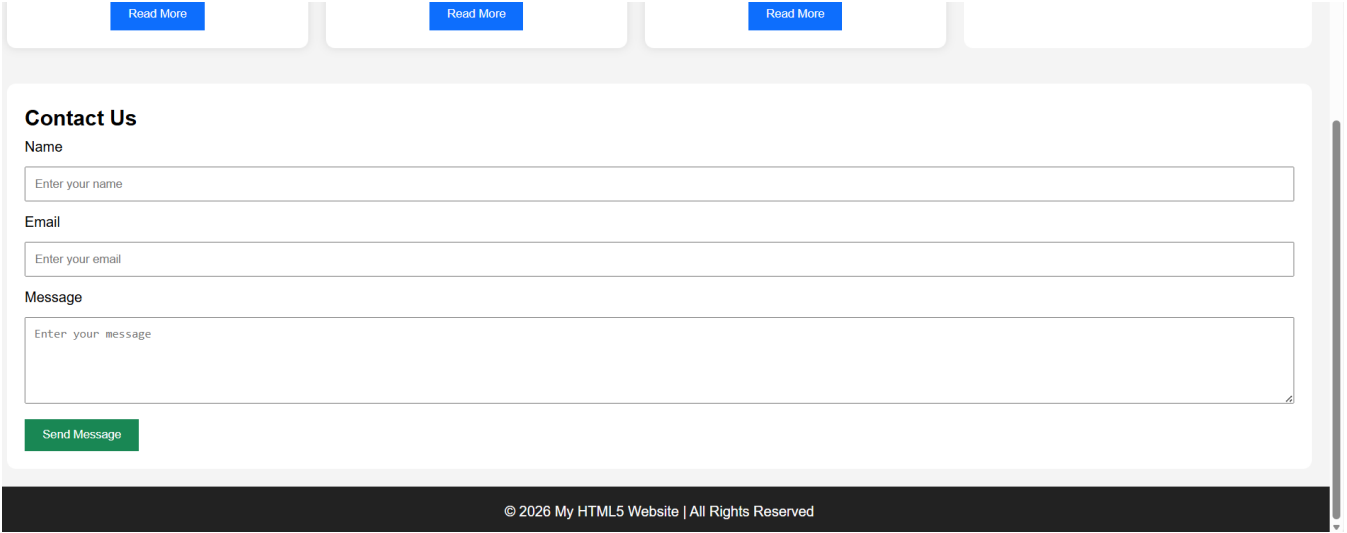
Output 1: Navigation Header & Section



Output 2: Responsive Flexbox Service Cards & News Sidebar



Output 3: Contact Form Controls & Footer Section



10. RESULT

Thus, a modern static webpage was successfully designed and developed using HTML5 semantic elements and CSS3 Flexbox layout concepts, and its layout responsiveness was validated against test cases.